MILITARY SPECIFICATION

TOWEL, PAPER (Disposable, Dental)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers two types of disposable towels suitable for protection of patient's clothing.

1.2 Classification. Disposable towels shall be of the following types, as specified, (see 6.1):

Type I. - Uncoated towel.
Type II. - Plastic coated towel.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

Federal

PPP-B-566 - Boxes, Folding, Paperboard
PPP-B-636 - Boxes, Shipping, Fiberboard.
PPP-B-676 - Boxes, Setup.

AMSC N/A FSC 8540

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Headquarters, Defense Personnel Support Center, Directorate of Medical Materiel, DPSC-ATT, 2800 South 20th Street Philadelphia, PA 19101, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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STANDARDS

Military

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.
MIL-STD-147 - Palletized Unit Loads.
MIL-STD-794 - Parts and Equipment, Procedures for Packaging of.

(Copies of specifications and standards required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issues of the documents which are DoD adopted shall be those listed in the issue of the DoDISS specified in the solicitation. Unless otherwise specified, the issues of the documents not listed in the DoDISS shall be the issue of the nongovernment documents which is current on the date of the solicitation.

TECHNICAL ASSOCIATION OF THE PULP AND PAPER INDUSTRY (TAPPI)

T 401 os - 74	Fiber Analysis of Paper and Paperboard
T 403 os - 76	Bursting Strength of Paper
T 410 os - 68	Weight Per Unit Area (Basis Weight or Substance) of Paper and Paperboard
T 432 su - 72	Water Absorbancy of Bibulous Papers
T 452 os - 58	Brightness of Paper and Paperboard
T 456 os - 68	Wet Tensile Breaking Strength of Paper and Paperboard

(Application for copies should be addressed to the Technical Association of the Pulp and Paper Industry, 1 Dunwoody Park, Atlanta, GA 30341).

(Nongovernment standards and other publications are normally available from the organization which prepare or which distribute the documents. These documents also may be available in or through libraries or other informational services.

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

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3. REQUIREMENTS

3.1 Material.

3.1.1 Tissue. Tissue shall be soft, facial quality material. Reclaimed fibers may be used in fabrication. Tissue shall be free of disagreeable odor and shall be in accordance with physical characteristics shown in table I when tested in accordance with table IV, except the wet tensile strength per 15 mm specimen width, in cross machine direction, shall be 0.04 kg minimum for Type II towel.

TABLE I. Physical properties of tissue.

Characteristic	Requirement
Fiber	100% bleached woodpulp
Bursting strength, 12 plies	15 points min
Basis weight, 500 single plies, 24 by 36 inches	10.5 lbs min
Water absorbency	10 seconds max
Brightness (white only)	80% min
Wet tensile strength per 15 mm	
Cross machine direction	.05 kg min
Both directions	.07 kg min

3.1.2 Plastic backing. Plastic backing (type II) shall be natural color, flexible polyethylene or polypropylene film. Film shall be at least 0.0004 inches thick.

3.2 Construction. The towel shall be not less than 19 inches in length and 13 inches in width. The towel shall be folded into four or eight panels.

3.2.1 Type I. Type I towel shall be fabricated from three or four plies of tissue.

3.2.2 Type II. Type II towel shall be fabricated of two or more plies of tissue bonded to a plastic backing. Minimum dimensions of backing shall be: length, length of towel minus 3/8 inch; width, width of towel minus 3/8 inch. Edge of backing and edge of towel shall be parallel. The distance from the edge of the towel to the edge of the backing shall be no more than 3/8 inch at any place.

3.3 Embossing. The towel shall be embossed to a depth and density which assure that tissue plies (type I) and tissue plies and backing (type II) adhere to one another and do not easily separate in use.

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3.3.1 Type I. Type I towel shall have an embossed pattern which covers the entire surface or each of the long edges and the center shall be embossed to form three parallel rows, each row a minimum of 1 inch in width.

3.3.2 Type II. For type II towel, embossed pattern shall cover the entire area which has plastic backing.

3.4 Color. Color of towel shall be white, light green, or light blue. Color shall be added to the tissue and not to the plastic backing (type II).

3.5 Workmanship. Towel shall be free of defects listed in table III and any other defects which may detract from its appearance or impair its serviceability.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Records. Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government, upon the Government's request, at any time, or from time to time, during the performance of the contract and for a period of 3 years after delivery of the supplies to which such records relate.

4.1.2 Inspection. Inspection, as used in this specification, is defined as both examination (such as visual and auditory without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item.

4.1.3 Certificates of quality. Certificates of quality, supplied by the manufacturer of the materials, may be furnished in lieu of actual performance of such testing by the contractor, provided lot identity has been maintained and can be demonstrated to the Government. The certificate shall include the name of the contractor, the contract number, the name of the manufacturer or supplier, the NSN, the Item Identification, the name of the material, the lot number, the lot size, the sample size, the date of testing, the test method, individual test results, and the specification requirements.

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4.2 Sampling.

4.2.1 For examination. Sampling for examination shall be in accordance with MIL-STD-105 and table II.

TABLE II. Sampling for examination.

Type of defect	Inspection level	AQL (percent defective)
For visual examination		
Major defects	II	4.0
Minor defects	II	6.5
For dimensional examination	S-2	2.5

4.2.2 For tests. Sampling for TAPPI tests indicated in table IV shall be as specified in TAPPI standards.

4.3 Examination. Towels shall be examined to determine compliance with all requirements in this specification. Examination shall be in accordance with, but shall not be limited to, defects listed in table III.

TABLE III. Classification of defects.

Category	Defects
Major	
101	Towel contains less than three plies of tissue (type I), or two plies of tissue (type II).
102	Tissue plies do not adhere to each other.
103	Tissue plies and backing do not adhere (type II).
104	Towel not soft and pliable.
105	Towel not free of dirt and grease.
106	Dry towel and towel saturated with water not free of disagreeable odor.
107	Towel not free of shives and slivers.
108	Towel not free of embedded foreign material.
109	Towel not free of tears and holes.
110	Backing not free of cracks and blisters (type II).
111	Edges not clean cut and well finished.
Minor	
201	Towel not uniform in color and finish.
202	Towel not free of wrinkles.
203	Towel not free of lint or dust.
204	Embossing pattern not symmetrical.

4.3.1 Dimensional examination. Towels shall be examined for defects in dimensions. Any dimension not within specification requirements shall be classified as a defect.

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4.4 Inspection of packaging. The sampling and inspection of the packing and marking for shipment and storage shall be in accordance with the quality assurance provisions of the applicable container specification and the marking requirements of MIL-STD-129.

4.5 Tests. Tests shall be conducted to determine compliance with specification requirements.

4.5.1 Tissue. Testing for physical characteristics of tissue, table I, shall be in accordance with table IV.

TABLE IV. Tests for physical characteristics of tissue.

Characteristic	TAPPI standards
Fiber identification	T 401 os - 74
Bursting strength, 12 plies	T 403 os - 76
Basic weight, 500 single plies, 24 by 36 inches	T 410 os - 68
Water absorbency	T 432 su - 72
Brightness (white only)	T 452 os - 58
Wet tensile strength per 15 mm	T 456 os - 68

5. PACKAGING

5.1 Preservation. Preservation shall be level A or C, as specified (see 6.1).

5.1.1 Level A.

5.1.1.1 Unit package. Five hundred towels of the type specified shall be packaged in a box conforming to PPP-B-566, PPP-B-636, class domestic, PPP-B-676 or in a paperboard dispenser type box. The box shall be such a size to keep the towels neatly and orderly stacked and to permit ease of dispensing one towel at a time. Closure of the box shall be effected by glueing or taping.

5.1.2 Level C. The level C preservation for towels shall conform to the MIL-STD-794 requirements for this level.

5.2 Packing. Packing shall be level A, B or C, as specified (see 6.1).

5.2.1 Level A. Two unit packages containing towels of one type only shall be packed in an exterior container designed for a type 2 load and conforming to PPP-B-636, class weather-resistant. Closure, strapping and waterproofing shall be as specified in the box specification.

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5.2.2 Level B. Two unit packages containing towels of one type only shall be packed in an exterior container designed for a type 2 load and conforming to PPP-B-636, class domestic. Closure shall be as specified in the box specification.

5.2.3 Level C. The level C packing for towels shall conform to the MIL-STD-794 requirements for this level.

5.2.4 Packing variation permitted. If the required number of units to be shipped is less than the number of units specified to be overpacked in an exterior container, such units shall be packed in an exterior container of suitable size and design, acceptable to a common carrier, which will insure safe delivery to destination.

5.2.5 Unitized loads. Unitized loads, commensurate with the level of packing specified in the contract or order, shall be used whenever total quantities for shipment to one destination equal 40 cubic feet or more. Loads shall be unitized on pallets conforming to MIL-STD-147. Quantities less than 40 cubic feet need not be unitized. Pallet loads including the pallet shall not exceed 43 inches in height, 48 inches in width, and 40 inches in length, and shall conform to Type IV or Type V.

5.2.5.1 Levels A and B. Towels packed as specified in 5.2.1 and 5.2.2 shall be unitized on pallets conforming to MIL-STD-147.

5.2.5.2 Level C. Towels packed as specified in 5.2.3 shall be unitized as specified in 5.2.5.

5.3 Marking.

5.3.1 Levels A, B, and C. Each unit pack, intermediate package, exterior container and unitized load shall be marked as specified in MIL-STD-129.

5.4 General.

5.4.1 Exterior container. Exterior container (see 5.2.1, 5.2.2, and 5.2.3) shall be of minimum tare and cube consistent with the protection required and shall contain equal quantities of identical stock numbered items to the greatest extent practicable.

5.4.2 Packaging inspection. The inspection of these packaging requirements shall be in accordance with 4.4.

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6. NOTES

6.1 Ordering data.

6.1.1 Acquisition requirements. Acquisition documents should specify the following:

(a) Title, number and date of this specification.
(b) Type required (see 1.2).
(c) Selection of applicable levels of packaging and packing (see 5.1 and 5.2).

6.2 National Stock Number (NSN) coverage. This specification covers the following item appearing in the Federal Supply Catalog:

Stock Number	Item Identification
8540-00-140-7317	TOWEL, PAPER, Plastic Coated on One Side, Disposable Dental, 500s
8540-00-140-7318	TOWEL, PAPER, Disposable, Dental, 500s

6.3 Limitations. This specification does not cover all types, classes, grades, styles, or sizes of the commodity indicated by the title of this specification, or those which are commercially available, but is intended to cover those items which are normally procured to meet Military requirements.

6.4 Supersession data. This specification includes the requirements of Specification Data Sheet No. 2, dated 9 July 1979, for NSNs: 8540-00-140-7317 and 8540-00-140-7318, Towels, Paper, Disposable, Dental.

MILITARY INTERESTS:

Custodians:

Army - MD
Navy - MS
Air-Force - 03

Review Activities:

Army - MD
Navy - MS
Air Force - 03

Civil Agency Coordinating Agency:
VA-OSS

05186

PREPARING ACTIVITY:

DoD-MB

AGENT:

DLA-DM

Project No. 8540-0035

Location: NCR

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1 DOCUMENT NUMBER	2 DOCUMENT TITLE
MIL-T-37968	- TOWEL, PAPER (Disposable, Dental)

3a. NAME OF SUBMITTING ORGANIZATION	4 TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)* _____
b. ADDRESS *(Street, City, State, ZIP Code)*	

5 PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording

c. Reason/Rationale for Recommendation

6 REMARKS

7a NAME OF SUBMITTER *(Last, First, MI)* – Optional	b WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8 DATE OF SUBMISSION *(YYMMDD)*

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